OMANSHU BHATT

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Portfolio: https://omanshu-bhatt-portfolio.vercel.app/ **GitHub:** https://github.com/Omanshu09 **LinkedIn:** https://www.linkedin.com/in/omanshubhatt/

PROFESSIONAL SUMMARY

Computer Engineering graduate who builds AI-powered and full-stack software. I enjoy working across the stack, writing clean code, building reliable systems and turning a rough idea into something people can actually use. I pick up new tools quickly and like solving practical problems. Interested in software and AI engineering roles where I can keep learning and ship real products.

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, SQL

Web & Frameworks: React, Next.js, Express.js, FastAPI, REST APIs, Node.js, Django, HTML, CSS, TypeScript, Flask, Microservices

Data & Databases: Elasticsearch, MySQL, PostgreSQL, MongoDB,Redis, Neon

AI / ML: PyTorch, LangChain, Generative AI, AI Agents, RAG, LLMs, Vector Databases, Prompt Engineering

DevOps & Systems: Docker, Kubernetes, Kafka, AWS, Git, GitHub, CI/CD (Jenkins, GitHub Actions), Vercel, Render

Distributed & Realtime Systems: Yjs, CRDTs, Ably, IndexedDB, WebRTC

Testing & QA: Manual Testing, Unit Testing, Integration Testing, API Testing, Postman, Pytest, Jest, SDLC, STLC

Core CS: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming

FEATURED PROJECTS

Web Research & Evidence Engine

- Built a web research engine for multi-source search, content extraction, relevance filtering, ranking and evidence-based answer generation using LLMs. Benchmarked across 100+ research queries, presented at the Agents for Humans Hackathon | Amazon/AWS
- **Tech:** Python, FastAPI, React, Web Search, Information Retrieval, BM25, RAG, LLMs, Docker

Code: https://github.com/Omanshu09/Search-Engine **Live:** https://search-engine-frontend-xi.vercel.app/

Real-Time Collaborative Engineering Workspace

- Built a real-time collaborative workspace with multi-user CRDT-based editing, near-instant synchronization, independent Python code execution, shared output and offline state recovery.
- **Tech**: React, TypeScript, Vite, Node.js, Express, Yjs, CRDTs, Ably, IndexedDB, PostgreSQL, Neon, E2B, REST APIs, Vercel, Render

Code: https://github.com/Omanshu09/Nexus-deployment **Live:** https://nexus-deployment-frontend.vercel.app/

Real-Time Translated Video Calling Platform

- Built a video calling platform that enables participants to speak different languages while seeing and hearing real-time translations, using Whisper STT and NLLB-200 entirely on-device via WebGPU with a WASM fallback.
- **Tech:** React, TypeScript, Vite, WebRTC, Web Workers, Whisper STT, NLLB-200, WebGPU, Web Speech API, Vercel

Code: https://github.com/Omanshu09/Fluo-Call **Live:** https://fluo-call.vercel.app/

AI Agents: Asha (Voice KAM), TaskPilot (Document Agent)

- Asha: voice-first AI account manager that diagnoses a seller's sales drop through conversation. Deterministic logic computes the metrics and picks each question, and the LLM only phrases replies, with validation that blocks invented numbers.
- TaskPilot: autonomous AI agent that turns a plain-language request into a validated Word document, it plans its own tasks, runs them in dependency order and stays reliable through plan validation, retries, rate limiting, execution logs and error handling.
- Tech: Python, FastAPI, Groq API, LLMs, Next.js, Node.js, Express, React, Next.js, TypeScript, python-docx, Pytest, Docker, Vercel, Render

Asha: https://ai-voice-omanshu.vercel.app/ **TaskPilot:** https://taskpilot-omanshu.vercel.app/

EXPERIENCE

Trainee Intern | Indian Oil Corporation Limited (IOCL) | July 2025 – August 2025

- Worked with operational data and database systems to analyze trends and recurring patterns in engineering workflows.
- Collaborated with IOCL engineers and technical teams, gaining experience with database management, enterprise engineering processes and quality-focused problem solving.

Virtual Intern — AI/ML & Robotics | Kodacy | July 2024 – August 2024

- Applied Python to AI/ML and data-analysis tasks, working with datasets to identify patterns, evaluate results and develop practical solutions.
- Developed Python-based robotics simulation prototypes involving algorithmic problem solving and automation.

CERTIFICATIONS and VIRTUAL EXPERIENCES

- Oracle — Agentic AI Certified Foundations Associate
- TCS iON Career Edge - Young professional
- University of Oxford — AI, Justice and rule of law
- Cisco — Data Science
- Commonwealth Bank — Software Engineering Job Simulation
- TATA — Cybersecurity Analyst
- BCGX — GenAI Job Simulation
- Deloitte — Cyber Job Simulation

EDUCATION

- **Govind Ballabh Pant University of Agriculture and Technology, Pantnagar | 2022 – 2026 | CGPA: 7.2**
  Bachelor of Technology (B.Tech) - Computer Engineering
- **Beersheba Senior Secondary School, Almora**
  Class XII | 2021 – 2022 | 90.4%
  Class X | 2019 – 2020 | 95.3%